

Crime and heat

Source-validation supplement

San Diego area | 24 September 2026

Independent administrative benchmarks, reported-offense data and observed weather extend the material available for this study. This supplement documents new evidence and its limits. It accompanies the published analysis and domestic-violence report; it does not replace their fitted estimates.

A documented reporting gap

California DOJ flags incomplete San Diego County Sheriff reporting for **November-December 2024 and January-June 2025** in both its arrest and DV-call datasets. The primary analyses exclude 2025 but retain the two flagged 2024 months. Their interpretation therefore requires this additional reporting qualification. The original release retains a separately labeled exploratory July-December 2025 DV model. [1, 2]

The notice establishes incomplete DOJ submissions. It does not identify the supplied export's missing rows, explain the reporting mechanism or establish a completeness percentage. The Sheriff-named reporting entity is also distinct from separately listed contract-city entities. Their operational overlap requires clarification.

Why row presence is insufficient

The Sheriff-named DV series has a row in every month of 2025 despite the official qualification. January-May contain 8, 8, 10, 6 and 12 reported calls. These values cannot be interpreted as complete activity or substituted for daily arrest counts. Calls resulting in agency reports include situations both with and without arrest. [2]

Implication for the analysis

A retrospective sensitivity excluding November-December 2024 could assess dependence on that interval. It must be specified and reported independently of whether its result crosses 0.05. Such a comparison would not recover missing records or establish completeness in earlier months. No new effect model is fitted in this source-acquisition update.

Source detail: [DOJ benchmark audit](#)

Additional outcome data

Source	Acquired material	Interpretation
California DOJ	1980-2025 county arrest/citation totals. 2001-2025 agency-month DV reports.	46 arrest years; 8,038 San Diego DV agency-month rows. Different units; reporting qualifications retained.
SANDAG / ARJIS	Group A and B public CIBRS snapshots, 2021-September 2026.	Group A: 624,649 agency-scoped incidents, including 66,681 DV-flagged incidents. Public-release completeness unverified.
San Diego Police	Seven annual NIBRS files, 2020-September 2026.	549,408 reported offense rows. Different agency population from the supplied Sheriff-labeled exports.
SANDAG historical DV [11]	37 annual values, 1986-2022.	Downloaded years differ from the chart title. Historical context; no daily heat inference.

Counting rules matter

SANDAG Group A uses a recorded DV indicator. This is broader in concept than selecting particular arrest-charge codes; it does not establish an arrest, unique victim or independently verified offense. The Sheriff subset contains 133,003 incident identifiers, including 15,513 DV-flagged identifiers, through 22 September 2026. Agency-specific date coverage varies. [3]

Group B has 112,170 rows, but its DV field is missing throughout the acquired snapshot. It cannot support a DV-arrest outcome. An incident can recur across arrest dates, so daily incident counts in Group B must not be summed as unique full-period incidents. [3]

SDPD supplies documented offense identifiers and occurrence dates. Distinct offenses reconcile across daily and monthly aggregates; case counts can overlap across offense categories. Its public CAD data exclude DV and other sensitive calls. A source with that exclusion cannot verify DV-call completeness. [4]

Neither continuous dates nor successful arithmetic reconciliation proves complete reporting. The 2026 files cover a partial year and remain subject to revision. Calls, offense records, arrest/citation totals and supplied source-ID groups remain separate series.

Observed weather and geography

Network	Acquisition and usable coverage
NOAA daily / GHCN	29 station archives; 1991-2025 extract. 25 stations satisfy at least 90% valid daily-maximum coverage during 2021-2024.
NOAA subdaily / ISD	12 stations; 379,574 quality-accepted hourly samples during 2021-2024. 13,235 complete civil station-days; five stations satisfy the 90% complete-day spatial-matching rule.
Historical baseline	Six stations meet the specified coverage criteria for May-September 1991-2020. Station-specific 90th/95th percentiles precede the primary 2021-2024 period.

Daily maximum coverage

Among eligible daily stations, the nearest station is selected before applying distance of at most 25 km and elevation difference of at most 200 m. The expanded network matches 94 of 112 ZIPs, yielding 136,243 paired ZIP-days. Within the corrected 64-ZIP sample, 54 ZIPs have matches. No second station is selected merely because the nearest fails the elevation rule. [5]

Measurement distinctions

GHCN maxima and minima follow station observation-day conventions. A reported daily average, where available, remains distinct from the high/low midpoint. ISD observations use their actual UTC timestamps and America/Los_Angeles civil dates. One accepted sample per hour defines sampled high, mean and low; incomplete civil days do not receive primary summary temperatures. Hourly extrema can miss short-lived peaks. [5, 6]

The 1991-2020 thresholds describe warm-season station distributions, not calendar-day percentiles or official heat alerts. Different stations and model grids measure different spatial supports. Pooled ZIP comparisons reuse station observations and must not be treated as independent measurement replicates.

Vista and San Marcos

The 2020 Census records 98,381 residents in Vista and 94,833 in San Marcos. Population size alone does not explain the large difference in supplied totals. Both contract with the Sheriff, and station coverage includes nearby unincorporated areas. The ambiguous recorded locality cannot support a municipal crime-risk denominator. The cause of the difference remains unresolved. [7, 8]

Evidence, verification and sources

Relationship to published research

A New Orleans study reported higher odds of DV-related calls during sustained extreme heat; a Los Angeles study found neighborhood variation in heat-associated crime. These support further investigation. Different outcomes, exposure contrasts and climates prevent treating either as replication of this arrest-record study. A positive estimate elsewhere does not resolve local reporting uncertainty. [9, 10]

Verification and remaining requirements

Independent calculations reconcile local data partitions and DOJ monthly/annual totals, check quality flags and unique station-day keys, and reproduce every complete ISD daily high, mean and low. All 23-, 24- and 25-hour civil-day boundaries are checked. These tests validate processing, not source ascertainment. Retrieval URLs, hashes, source dictionaries, separate aggregate series and code accompany the release.

Further models require explicit counting and geographic units, reporting exclusions, exposure definitions, lags and multiplicity rules before inspecting their estimates. The agency export dictionary and extraction crosswalk remain unavailable. No missing daily counts are imputed; no sources are pooled as equivalent crime events.

References

1. California DOJ. Arrests/Arrest Dispositions context. Revised June 2026; p. 7.
 2. California DOJ. Domestic Violence Related Calls for Service context. Revised June 2026; pp. 1, 5.
 3. SANDAG / ARJIS. Public CIBRS Group A. Dataset ID pr74-d3tr. Group B: huzf-mi2z.
 4. City of San Diego. Police NIBRS offense data and source definitions. CAD exclusions.
 5. NOAA NCEI. Global Historical Climatology Network-Daily documentation.
 6. NOAA NCEI. Integrated Surface Database format and quality documentation.
 7. U.S. Census Bureau. California incorporated places: 2020 Census population.
 8. San Diego County Sheriff. Service jurisdictions and contract cities.
 9. Dey AK et al. Extreme Heat and Calls to Law Enforcement Related to Domestic Violence. *JAMA Netw Open*. 2025;8:e2530530.
 10. Heilmann K, Kahn ME, Tang CK. The urban crime and heat gradient in high and low poverty areas. *J Public Econ*. 2021;197:104408.
 11. SANDAG. Historical county domestic-violence table; acquired values 1986-2022.
- Study and complete source audits: boommerrawlings.com/writing/data-analysis/crime-and-heat/